

Assignment 2: Design Your Dream Logo

Logo Name: **Zenith Orbit**

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Concept

Zenith Orbit is an imaginary technology company. The logo is a radiating sun/orbit emblem built from two concentric circles, eight symmetric rays, a curved base arc, a small crown accent, and a wavy underline. The design is built around a single **vertical mirror axis** ($x = 0$).

Equations Used (16 total, exactly as built in Desmos)

#	Equation (as entered in Desmos)	Type	Represents
1	$x^2 + y^2 = 20.25$	Circle	Outer ring boundary
2	$x^2 + y^2 = 5.35$	Circle	Core boundary
3	$x^2 + y^2 \leq 5.29$	Circle (ineq.)	Core fill (color)
4	$x^2 + y^2 \leq 10.24 \{x^2 + y^2 \geq 5.29\}$	Circle (ineq.)	Ring band fill (color)
5	$y = x, \{1.8 \leq x \leq 2.8\}$	Linear	Ray, upper-right
6	$y = -x, \{1.8 \leq x \leq 2.8\}$	Linear	Ray, lower-right
7	$y = x, \{-2.8 \leq x \leq -1.8\}$	Linear	Ray, lower-left
8	$y = -x, \{-2.8 \leq x \leq -1.8\}$	Linear	Ray, upper-left
9	$x = 0, \{2.3 \leq y \leq 3.6\}$	Linear	Ray, top
10	$x = 0, \{-3.6 \leq y \leq -2.3\}$	Linear	Ray, bottom
11	$y = 0, \{2.3 \leq x \leq 3.6\}$	Linear	Ray, right
12	$y = 0, \{-3.6 \leq x \leq -2.3\}$	Linear	Ray, left
13	$y = -0.09x^2 - 4.35, \{-3 \leq x \leq 3\}$	Quadratic	Base arc
14	$y = x + 7.1, \{-0.8 \leq x \leq 0\}$	Linear	Crown, left side
15	$y = -x + 7.1, \{0 \leq x \leq 0.8\}$	Linear	Crown, right side (mirror of #14)
16	$y = 0.10 \sin(2x) - 6.2, \{-3 \leq x \leq 3\}$	Trigonometric	Wavy underline

Function types present: circle, linear, quadratic, and trigonometric (4 types). *Note: this set does not currently include an absolute value equation, so it falls one short of a 5-type minimum if that is required — an easy fix is adding one small absolute-value crown or accent curve.* Inequalities (#3, #4) shade the core and ring band with color. Symmetry is clearly visible: rays #5/#8 and #6/#7 are mirror pairs across the vertical axis, rays #11/#12 mirror each other, and the crown lines #14/#15 form a symmetric peak. Both circles (#1, #2) and the quadratic base arc (#13) are automatically symmetric about the y -axis since they contain only x^2 terms.

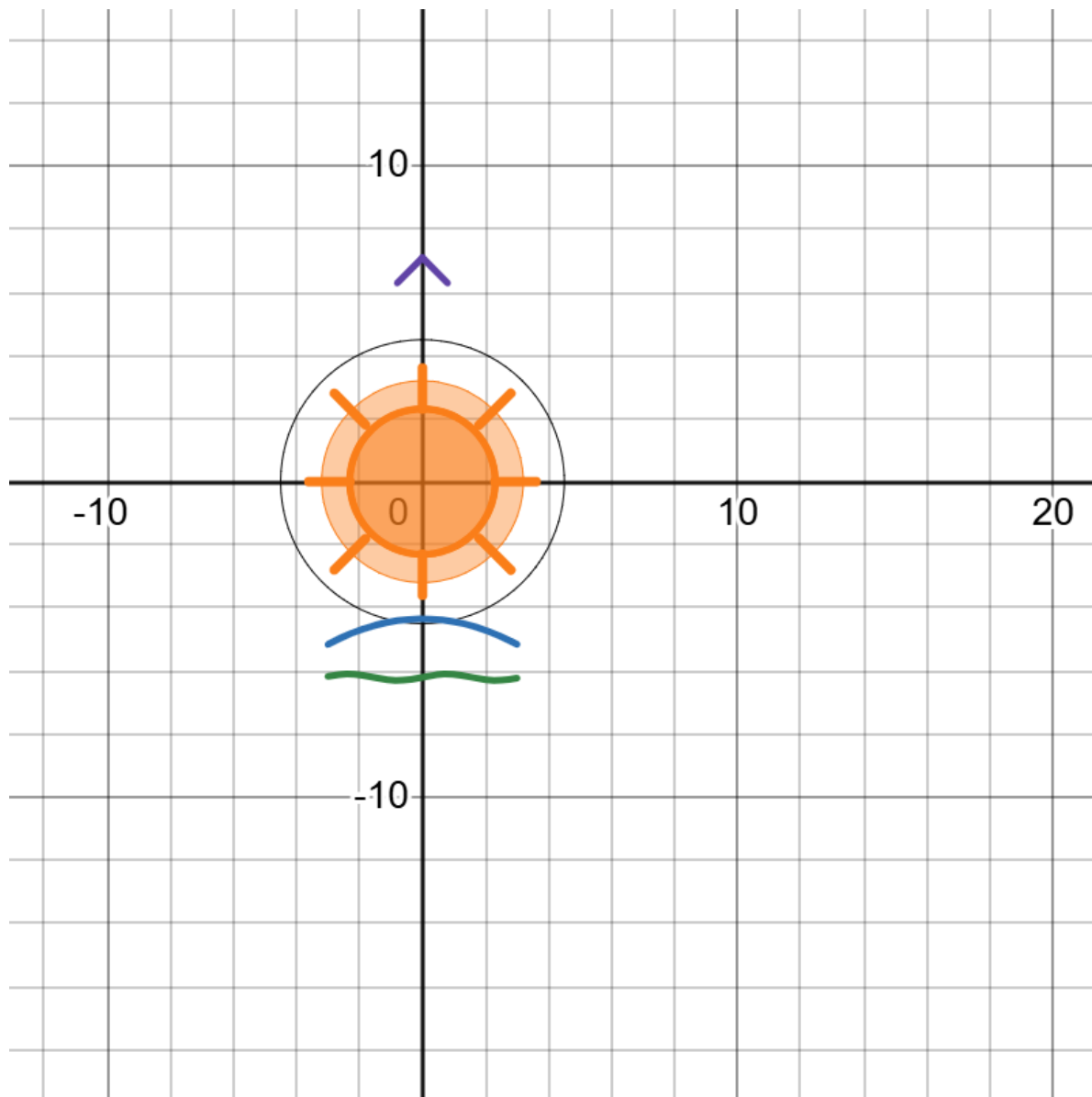
Mathematics Behind the Design

The logo is anchored by two concentric circles, $x^2 + y^2 = 20.25$ and $x^2 + y^2 = 5.35$, which are inherently symmetric under any rotation since they depend only on $x^2 + y^2$; the inequalities #3 and #4 shade the core and the ring band between them without needing any extra curves. The eight rays come from just four lines – $y = x$, $y = -x$, $x = 0$, and $y = 0$ – each restricted to a short interval on both sides of the origin, so domain restrictions turn four infinite lines into eight finite, evenly spaced spokes. The crown was built from two ordinary linear pieces, $y = x + 7.1$ and $y = -x + 7.1$, placed on mirrored domains so they meet at a single peak and form a V-shape without using an absolute value function. The quadratic base arc, $y = -0.09x^2 - 4.35$, curves gently because its coefficient is small and negative, and being a pure x^2 expression it is automatically symmetric. The sine wave underline adds a repeating, flowing accent that none of the other function types could produce on their own.

Explanation

This logo uses four different types of mathematical functions: linear, quadratic, circular, and trigonometric functions. Circle equations formed the outer ring and core, linear equations built all eight rays, the two-sided crown, and were restricted with domains to keep them as short segments instead of infinite lines, the quadratic function created the curved base arc, and a sine function produced the wavy underline. Inequalities were used on the two circle equations to shade the core and the ring band with distinct colors. The main challenge was getting the domain restrictions on each ray to line up exactly with the circle radii, so the rays appeared to start right at the edge of the ring instead of overlapping it or leaving a gap. I also had to adjust the crown's two linear pieces carefully so they met at a single point. During this assignment, I learned how restricting the domain of ordinary lines can produce complex radial symmetry, and how inequalities can shade regions between two curves.

Screenshot of the Desmos Graph



Desmos Graph Link: <https://www.desmos.com/calculator/n59iwstno2>